The self-coupled dirac16complex in gravitational fields: provenance

Sources, decisions, derivations and the verification record of the nine-channel self-coupling, the Fierz matrix, the covariant nonlinear field equation and the Kohn–Sham theory of dirac16complex in the three epochs — sections 15–16 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states what was asked, what was used, every decision taken with the equation behind it, how each result was derived and how it was checked: the 24 checks of sections 15–16 of 7+1.nb (24 True), the 29 boolean checks of the independent Python program tools/ref_ks_gravity.py (29 True), the comparison of every number printed by both programs, and the SHA-256 hashes of the files.

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1. The task

This stage of the project asked (summarized; the original wording is the author's private working material):

1. a GENERALIZED self-coupling of the 16-component field dirac16complex, beyond the four contact couplings (scalar, pseudoscalar, vector, axial) of the earlier Kohn–Sham theory of the notebook;
2. the Kohn–Sham theory of dirac16complex in GRAVITATIONAL fields, with the canonical spin connection coupling gravity to the field, in the primordial epoch (near $x_0 = 0$), the modern epoch and the far future;
3. everything on the 7+1 spacetime of this project — one time coordinate x_0, seven space coordinates $x_1, \ldots, x_7$ — with the ideas (not the 4+4 metric) of the repository github.com/once-ere/Dirac_claude;
4. for every documentation file a companion provenance file and a symbol file, in Markdown and LaTeX with PDFs; no document refers the reader to another one; every equation used is written in the document that uses it;
5. and (27 October 2026) to make clear in all documentation, examples and provenance records that the xAct list $\{7, 1, 0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1, -1, \ldots, -1)$ is simply the squares of the octonion basis (section 1.1).

1.1 The signature: the author's explanation of 27 October 2026

Source. On 27 October 2026 the author of the project explained that the xAct list $\{7,1,0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1,-1,\ldots,-1)$ is simply the squares of the octonion basis, and asked that this be made clear in all documentation, examples and provenance records. **What it corrected.** The version of 7+1.nb pushed in the commits 52d8729 to dc2f178 of github.com/once-ere/7plus1 declared the xAct metric of section 12 with `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of η; since then section 12 declares `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`, exactly as the author's notebook Einstein-Lovelock-7+1-Nash.nb does. **What was added.** Checks in 7+1.nb — section 4.7: η is the squares of the octonion basis, $e_A e_A = \eta_{AA} e_0$ and $\mathrm{Re}(e_A e_B) = \eta_{AB}$; section 12.2: `SignatureOfMetric[g71]` $= \{7,1,0\}$ with `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$, and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$; section 17: the flat metric $\mathrm{diag}(+1,-1,-1,-1)$ of the world brane is the squares of the basis e_0, e_1, e_2, e_3 of the quaternion subalgebra — and the same statements in the Python programs `tools/ref71.py`, `tools/ref_gravity.py` and `tools/ref_sm.py`; the explanation itself in the text of 7+1.nb (sections 1.2, 1.4, 4.2, 4.7, 7, 11, 12.2, 13, 15.1, 17) and in every document, symbol file and provenance file of the repository. **What did not change.** No result. Compared output by output with the backup below, every output of 7+1.nb is the same, except the declared list itself, the new checks and the check counts (and, as in every run, the name of the private context and the measured times); in particular every component of every Lovelock tensor is unchanged. The components follow from the component matrix of the metric, and the sign of the determinant, `SignDetOfMetric`, is -1 for both lists. **Backup.** The evaluated notebook as it was before is kept in the workspace as `backups/7+1_2026-10-27_before_signature.nb` (the folder backups/ is excluded from the repository by .gitignore), 1034401 bytes, SHA-256 `060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`; `tools/verify_7p1_nb.py` compares every input, text and section cell of 7+1.nb with it and finds every cell present, each one unchanged or changed by this explanation, and it compares every output in the same way.

The explanation, in full. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2. Inputs

2.1 Objects of the earlier sections of 7+1.nb

Used unchanged: the real Dirac matrices `Γ17[A]`, `Γ17up[A]`, the metric `η17`, the chirality `χ17` $= i\,\Gamma_0\Gamma_1\cdots\Gamma_7$, the chiral basis `EL`, `ER`, the Hamiltonian matrices `αD[j]`, `βD`, the curvature engine and the frame data of the cosmological metric (`fdCos`), the closed forms of the Lovelock tensors of the cosmological metric (`ECos`, with the rewriting `toH` in terms of H_a, H_b, H_a', H_b'), and the Hartree couplings and saturation state of the uniform

matter of section 8.7 of the notebook:

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \quad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \quad \chi = i\,\Gamma_0\Gamma_1\cdots\Gamma_7,$$

$$E_L = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1_8 \\ -i1_8 \end{pmatrix}, \quad E_R = \tfrac{1}{\sqrt{2}} \begin{pmatrix} -i1_8 \\ 1_8 \end{pmatrix},$$

$$G_S = -100000, \quad G_V = 10000 \quad (m = 1),$$

$$k_{F,\mathrm{sat}} = 0.75663615070149289469\ldots \quad \text{(where } E/A \text{ is minimal at } a = b = 1\text{)}.$$

2.2 Software

- Wolfram Language 15.0.1 (front end WolframNB.exe), with the standard package NumericalDifferentialEquationAnalysis (its function `GaussianQuadratureWeights`) for the Gauss–Legendre nodes.
- Python 3.13.17 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0 (the independent program).
- Textbook results, each derived and checked here: Fierz rearrangement (M. Fierz, Z. Phys. 104 (1937) 553), Hohenberg–Kohn and Kohn–Sham (Phys. Rev. 136 (1964) B864; 140 (1965) A1133), relativistic mean-field matter (J. D. Walecka, Ann. Phys. 83 (1974) 491).

3. Decisions

#	decision	reason / equation
1	basis of the 256 matrices: $\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}$, $\sigma_r = \frac{r(r-1)}{2} \bmod 2$	$(\Gamma_0\Gamma^{A_1}\cdots\Gamma^{A_r})^\dagger = (-1)^{r(r-1)/2}\,\Gamma_0\Gamma^{A_1}\cdots\Gamma^{A_r}$ (reversing r distinct anticommuting factors); the phase i^{σ_r} makes $\Gamma_0\Theta^K$ Hermitian, so every bilinear is real (section 4)
2	the self-coupling is organized by rank, one coupling g_r per rank	each rank is a Lorentz-invariant class (the generators $\frac{1}{4}[\Gamma^A,\Gamma^B]$ map products of rank r to rank r)
3	the four couplings of section 8 are identified as $g_0 = g_S$, $g_1 = g_V$, $g_7 = -g_A$, $g_8 = -g_P$	$I_8 = -(\bar\psi i\chi\psi)^2$ and $I_7 = -(\bar\psi\Gamma^A\chi\psi)(\bar\psi\Gamma_A\chi\psi)$
4	exchange is folded into the direct terms by the Fierz matrix	$E_x = -\sum_{r,s} \frac{g_r}{2} F_{rs} D_s$, hence $G = (1 - F^{\mathsf{T}})g$
5	uniform matter: spin-saturated, no Dirac sea, Hartree–Fock couplings $G_S = G_0$, $G_V = G_1$	only $D_0 = \rho_S^2$ and $D_1 = \rho^2$ are non-zero for $R = (\rho_S 1 + \rho\Gamma_0)/16$
6	comoving Fermi ball: the comoving momenta are conserved by a homogeneous expansion	the modes $e^{i(\vec p\cdot\vec x + \vec q\cdot\vec y)}$ keep $\vec p, \vec q$; the physical momenta are $\vec p/a$, $\vec q/b$
7	the comoving Fermi momentum is the flat-space saturation value	the matter starts (at $a = b = 1$) in its ground state at zero pressure
8	adiabatic (instantaneous) Kohn–Sham state	valid when $H_a, H_b \ll$ orbital energies; the pair creation of the epochs measures the departures
9	Gauss–Legendre quadrature with 96×64 nodes in (k, θ)	smooth integrands; the flat limit reproduces the closed forms of section 8.7 to 10^{-11}
10	the 00 equation is imposed on the initial data only	it is a constraint without second derivatives; its preservation is a check of the whole system
11	NDSolve with explicit Runge–Kutta of order 8 and InterpolationOrder → All	without it the interpolation between steps (order 3) showed an apparent constraint violation of 2×10^{-6}

4. Derivations

Reality of the bilinears. $\Gamma_0\Gamma^{A\dagger}\Gamma_0 = \Gamma^A$, so $(\Gamma_0\Gamma^{A_1}\cdots\Gamma^{A_r})^\dagger = \Gamma^{A_r\dagger}\cdots\Gamma^{A_1\dagger}\Gamma_0 = \Gamma_0\Gamma^{A_r}\cdots\Gamma^{A_1} = (-1)^{r(r-1)/2}\Gamma_0\Gamma^{A_1}\cdots\Gamma^{A_r}$ (distinct indices anticommute). Multiplying by i^{σ_r} with $\sigma_r = \frac{r(r-1)}{2} \bmod 2$ compensates the sign: $(i^{\sigma_r})^*(-1)^{r(r-1)/2} = i^{\sigma_r}$.

Fierz matrix. Completeness, $\sum_K (\Theta_K)_{cd}(\Theta^K)_{ab} = 16\delta_{ad}\delta_{bc}$, gives $\sum_{K\in r}(a\Theta_K b)(c\Theta^K d) = \sum_J C_{rJ}(a\Theta_J d)(c\Theta^J b)$ with $C_{rJ} = \frac{1}{256}\sum_{K\in r}\mathrm{tr}(\Theta_K\Theta_J\Theta^K\Theta^J)$. Moving Γ^A through a product of s distinct Γ's gives $(-1)^s$ if A is not among them and $(-1)^{s-1}$ if it is; for K of rank r sharing l indices with J the sign is $(-1)^{rs-l}$, and there are $\binom{s}{l}\binom{8-s}{r-l}$ such K:

$$F_{rs} = \frac{16}{256}\sum_l (-1)^{rs-l}\binom{s}{l}\binom{8-s}{r-l} = \frac{1}{16}\sum_l (-1)^{rs-l}\binom{s}{l}\binom{8-s}{r-l}.$$

Densities of uniform matter. A 16-component field has 8 positive-energy states per momentum, so $\rho = 8\int \frac{d^7k}{(2\pi)^7}$ over the Fermi ball (comoving, divided by the comoving volume factor a^3b^4). With $k_a = k\cos\theta$, $k_b = k\sin\theta$:

$$d^3k_a\, d^4k_b = (4\pi)(2\pi^2)\, k^6\cos^2\theta\sin^3\theta\, dk\, d\theta, \qquad \frac{8\cdot 8\pi^3}{(2\pi)^7} = \frac{1}{2\pi^4},$$

$$\int_0^{\pi/2}\cos^2\theta\sin^3\theta\, d\theta = \frac{2}{15}, \qquad \rho = \frac{k_F^7}{105\pi^4 a^3 b^4}.$$

The kinetic pressures are the momentum flux per direction: three ordinary directions share k_a^2/a^2, four auxiliary directions share k_b^2/b^2, which gives $p_a = \langle k^2\cos^2\theta/(3a^2\omega)\rangle + \Pi$ and $p_b = \langle k^2\sin^2\theta/(4b^2\omega)\rangle + \Pi$; the interaction pressure is $\Pi = \rho\,\partial\varepsilon_{\rm int}/\partial\rho - \varepsilon_{\rm int}$ of the mean-field energy $\varepsilon_{\rm int} = -\frac{(M-m)^2}{2G_S} + \frac{G_V}{2}\rho^2$ at fixed M.

Conservation. $\varepsilon(a,b,M)$ is stationary in M (that is the gap equation), so $\varepsilon' = \partial_a\varepsilon\, a' + \partial_b\varepsilon\, b'$; differentiating the comoving integrals ($\rho \propto a^{-3}b^{-4}$, $\partial_a\omega = -k^2\cos^2\theta/(a^3\omega)$) gives

$$\varepsilon' = -3H_a(\varepsilon + p_a) - 4H_b(\varepsilon + p_b).$$

Constraint preservation. The Lovelock tensors are divergence-free for every metric, $\nabla_\mu E^{(k)\mu}{}_\nu = 0$; for the cosmological metric the $\nu = 0$ component reads $\frac{d}{dx_0}E^{(k)0}{}_0 + 3H_a(E^{(k)0}{}_0 - E^{(k)1}{}_1) + 4H_b(E^{(k)0}{}_0 - E^{(k)4}{}_4) = 0$. With the 11 and 44 equations imposed and the conservation law above, the 00 equation therefore holds at all times if it holds initially.

5. Verification

Evaluation of 7+1.nb: Wolfram 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026); finished Tue 27 Oct 2026 19:31:35; message cells 0. The checks of sections 15–16:

#	check	result	section
1	256 multi-indices in 9 ranks with 1, 8, 28, 56, 70, 56, 28, 8, 1 members	True	15.1
2	every bilinear psibar Theta^K psi is real: Gamma17[0].Theta^K Hermitian for all 256 K	True	15.1
3	Theta_K.Theta^K == ID16 and Tr[Theta_K.Theta^J] == 16 delta_KJ	True	15.1
4	Hodge duality: Theta^K.omega17 == (phase) Theta^(complement of K), |phase| = 1	True	15.1
5	the couplings of section 8 are four of the nine: I_0 == (psibar psi)^2, I_1 == (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 == -(psibar i chi17 psi)^2, I_7 == -(axial invariant)	True	15.1
6	local Lorentz invariance: I_r(S psi, psibar S^-1) == I_r(psi, psibar) for r = 0..8 (random finite Lorentz transformation), and S^-1 == Gamma17[0].S^dagger.Gamma17[0]	True	15.1
7	Fierz matrix: closed form == (1/256) Sum_{K in r} Tr[Theta_K.Theta_J.Theta^K.Theta^J] for the first and last J of every rank s	True	15.2
8	F.F == IdentityMatrix[9]; eigenvalues: five +1 and four -1	True	15.2
9	Fierz identity for commuting spinors (random): Sum_{K in r} (a Theta_K b)(c Theta^K d) == Sum_s F_rs Sum_{J in s} (a Theta_J d)(c Theta^J b), r = 0..8	True	15.2
10	Hartree-Fock energy E_H + E_x == Sum_s (G_s/2) D_s with G = (1 - Transpose[F]) g (random density matrix and couplings)	True	15.2

#	check	result	section
11	uniform density matrix (rhoS ID16 + rho Gamma17[0])/16: D_0 == rhoS^2, D_1 == rho^2, D_2 .. D_8 == 0	True	15.2
12	section 8 recovered: with g = (gS, gV, 0, 0, 0, 0, 0, -gA, -gP) the exchange energy of uniform matter is that of section 8.7	True	15.2
13	the Hartree-Fock couplings of section 8.7: gS = -100000, gV = 10000 give G_0 = -98750, G_1 = 20000	True	15.2
14	nonlinear field equation: dL_self/d psibar_a == -Sum_r g_r Sum_K (psibar Theta_K psi) (Theta^K psi)_a	True	15.3
15	chi17 commutes with every generator (Gamma17up[A].Gamma17up[B] - Gamma17up[B].Gamma17up[A])/4, hence with every Omega_mu	True	15.4
16	curved chiral equations: E_L^dag.Gamma17[0].(i Gamma^mu D_mu - m) psi == (i d0 + i(3Ha/2 + 2Hb) + tauO.d) phi_L - m phi_R, and the same for E_R with -tauO.d	True	15.4
17	flat limit a = b = 1, kF = kFsat: m* == 0.3480973401... and E/A - m == -0.0283214279... (section 8.7), to 10^-11	True	16.1
18	flat limit: pa == pb == rho eps_F - eps (Gibbs-Duhem), which is 0 at the saturation density	True	16.1
19	energy-momentum conservation of the Kohn-Sham matter: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) == 0 (a = 1 + 0.3x + 0.05x^2, b = 1/(1 + 0.2x), x0 = 0.7; relative 10^-8)	True	16.1
20	primordial: rho x0 == kFsat^7/(105 Pi^4) at every x0, and m* decreases monotonically to below 0.05 at x0 = 10^-4	True	16.2
21	modern (b = 1): eps a^3 tends to a constant and pa/eps -> 0 (pressureless in ordinary space)	True	16.2
22	far future (de Sitter, H = 0.1): rho == rho0 e^{-3 H x0} and 1 - m* decays like rho	True	16.2
23	the 11 and 44 Einstein-Lovelock equations are linear in (Ha', Hb'); the 00 equation contains neither (constraint)	True	16.3
24	self-consistent evolution: the constraint Sum alpha_k E^(k)0_0 - Lambda == kappa eps is preserved to 10^-7 (relative) for Einstein and for Einstein-Lovelock (alpha2 = 20, alpha3 = 500)	True	16.3

The independent Python program tools/ref_ks_gravity.py — its boolean checks:

#	check	result
1	256 products in 9 ranks: 1, 8, 28, 56, 70, 56, 28, 8, 1	True
2	every bilinear psibar Theta^K psi is real: Gamma17[0] Theta^K Hermitian (Theta^K = i^((r(r-1)/2) mod 2) Gamma^K)	True
3	Theta_K Theta^K = 1_16 and tr(Theta_K Theta^J) = 16 delta_KJ	True
4	Fierz completeness: sum_K (Theta_K)_cd (Theta^K)_ab = 16 delta_ad delta_bc	True
5	Hodge duality: Theta^K omega17 = +- Theta^(complement of K) (up to a phase)	True
6	stage-2 channels: I_0 = (psibar psi)^2, I_1 = (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 = -(psibar i chi psi)^2, I_7 = -(axial)	True
7	local Lorentz invariance: I_r(S psi, psibar S^-1) = I_r(psi, psibar), r = 0..8 (random finite boost-rotation)	True
8	Fierz matrix F_rs = (1/256) sum_{K in r} tr(Theta_K Theta_J Theta^K Theta^J), independent of the representative J of rank s	True
9	Fierz matrix closed form F_rs = (1/16) sum_l (-1)^(rs-l) C(s,l) C(8-s,r-l)	True
10	F^2 = 1 (the Fierz transformation is an involution)	True
11	Fierz identity for commuting spinors: sum_{K in r} (a Theta_K b)(c Theta^K d) = sum_s F_rs sum_{J in s} (a Theta_J d)(c Theta^J b)	True
12	Hartree-Fock energy E_H + E_x = sum_s (G_s/2) D_s with G = (1 - F^T) g (random density matrix, random couplings)	True
13	uniform matter R = (rho_S + rho Gamma17[0])/16: only D_0 = rho_S^2, D_1 = rho^2; with g_0 = gS, g_1 = gV, g_7 = -gA, g_8 = -gP the exchange energy is that of stage 2 (section 8.7)	True
14	uniform density matrix: D_0 = rho_S^2, D_1 = rho^2, D_2..D_8 = 0	True
15	stage-2 couplings gS = -100000, gV = 10000: G_0 = -98750, G_1 = 20000 (the HF couplings of section 8.7)	True
16	nonlinear field equation: dL_int/d psibar_a = -sum_r g_r sum_K (psibar Theta_K psi)(Theta^K psi)_a	True
17	the chirality chi17 commutes with every generator Sigma^AB, hence with every spin connection Omega_mu	True
18	curved chiral equations: E_L^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) + tauO.d/(a,b)) phi_L - m phi_R	True
19	curved chiral equations: E_R^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) - tauO.d/(a,b)) phi_R - m phi_L	True
20	flat limit a = b = 1, kF = kF0 (stage-2 saturation): m* = 0.3480973401..., E/A - m = -0.0283214279...	True

#	check	result
21	flat limit: isotropic pressure pa = pb = rho eps_F - eps (Gibbs-Duhem, eps_F = V0 + sqrt(kF^2 + m*^2))	True
22	energy-momentum conservation eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for the KS matter (arbitrary a, b; relative residual)	True
23	primordial: rho(x0) x0 = kF0^7/(105 pi^4) (density grows like 1/x0)	True
24	primordial: m* decreases toward 0 as x0 -> 0 (blue-shifted ordinary momenta, attractive scalar coupling)	True
25	modern: eps a^3 -> constant and pa/eps -> 0 (dirac16complex behaves as pressureless matter in ordinary space)	True
26	far future: rho = rho0 e^{-3 H x0} and 1 - m* -> 0 like rho	True
27	the 00 closed forms of E^(1), E^(2), E^(3) (pair-set algorithm, tools/lovelock71.py) agree with the closed forms of section 12.6 of 7+1.nb	True
28	Einstein-Lovelock equations are linear in (Ha', Hb') (quasi-linear second-order system)	True
29	the constraint Sum alpha_k E^(k)0_0 - Lambda = kappa eps is preserved by the evolution (relative violation < 1e-7) in both cases	True

Numbers printed by both programs (relative differences):

quantity	values	max. relative difference
flat limit: m^*, ρ, ε	3	7.5×10^{-15}
primordial $m^*(x_0)$	6	5.3×10^{-15}
primordial p_a/p_b	3	2.3×10^{-15}
modern $m^*(a)$	5	4.1×10^{-15}
modern p_b/ε at $a = 100$	1	3.1×10^{-15}
far future $m^*(x_0)$	5	4.1×10^{-15}
Einstein run: $H_b(0)$; a, b, H_a, H_b at $x_0 = 20$	5	2.6×10^{-12}
Einstein–Lovelock run: $H_b(0)$; a, b, H_a, H_b at $x_0 = 20$	5	2.1×10^{-12}

Checks made while generating the documents: 16 F printed by 7+1.nb == closed form: True.

SHA-256 of the files (at the time this document was generated):

```
7+1.nb                          1314b1e09e6b6eefbd9ee5974e19d186a5e996907b2eb68ddb6f23aae85aa
    ...aa5
tools/make_7p1_nb.py            354a0210e60510499753c2cf91274850cedef502d6f705efc410c5a8bbae7
    ...b2d
tools/nb_part_gravity.py        c4caf2ad5c932e8f603341edade5f4b98fa784feb05906c051eab7c884cd6
    ...8f8
tools/nb_part_ks_gravity.py     a656c0ad040b4576b75b17e029545109e531c8e656ad99bfdb5cba946e967
    ...772
tools/nb_part_sm.py             43e92beed57453a4b57931de6d603ac6c218efba6e4005133b69b8054a975
    ...c5b
tools/nb_part_coupled.py        49c61aeb6affe14ebbcccb32d2069504b9fc035313ea2fabbd0a3392a9613
    ...3f5
tools/lovelock71.py             d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af9
    ...5a1
tools/ref_gravity.py            f07cf8adaff585af89c87d19532f1aaa7db606ad88ae90483b2d1f41341ed
    ...f36
tools/ref_ks_gravity.py         b6488ef24b1c16aacd8ff6cf6a5a1ba3f316e082449ad97480022d1a26383
    ...27f
tools/ref_sm.py                 f3f80806893f42bbf166e1f1de896ffd611dbfe847685da8e3ee367e9cde6
    ...aba
tools/ref_coupled.py            695dbd9375753653472d9c37502f0067c4a64fe01a64ba8c05e8b2c5da85a
    ...a58
tools/ref71.py                  1a7a06eb0add3f459611776dc9c474a5bb959d947a9db949db4612b1ab943
    ...4e0
tools/ref44.py                  ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064f
    ...af5
```

Output of tools/ref_ks_gravity.py (independent checks of sections 15-16; exit code 0) (complete):

```
256 products in 9 ranks: 1, 8, 28, 56, 70, 56, 28, 8, 1 : True
every bilinear psibar Theta^K psi is real: Gamma17[0] Theta^K Hermitian (Theta^K =
    ...i^((r(r-1)/2) mod 2) Gamma^K) : True
Theta_K Theta^K = 1_16 and tr(Theta_K Theta^J) = 16 delta_KJ : True
Fierz completeness: sum_K (Theta_K)_cd (Theta^K)_ab = 16 delta_ad delta_bc : True
Hodge duality: Theta^K omega17 = +- Theta^(complement of K) (up to a phase) : True
stage-2 channels: I_0 = (psibar psi)^2, I_1 = (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 =
    ...-(psibar i chi psi)^2, I_7 = -(axial) : True
local Lorentz invariance: I_r(S psi, psibar S^-1) = I_r(psi, psibar), r = 0..8 (random finite
    ...boost-rotation) : True
Fierz matrix F_rs = (1/256) sum_{K in r} tr(Theta_K Theta_J Theta^K Theta^J), independent of
    ...the representative J of rank s : True
Fierz matrix closed form F_rs = (1/16) sum_l (-1)^(rs-l) C(s,l) C(8-s,r-l) : True
F^2 = 1 (the Fierz transformation is an involution) : True
eigenvalues of F: number of +1, number of -1 : (5, 4)
Fierz matrix (rows r, columns s) times 16 : [[1, 1, 1, 1, 1, 1, 1, 1, 1], [8, -6, 4, -2, 0, 2,
    ...-4, 6, -8], [28, 14, 4, -2, -4, -2, 4, 14, 28], [56, -14, -4, 6, 0, -6, 4, 14, -56],
    ...[70, 0, -10, 0, 6, 0, -10, 0, 70], [56, 14, -4, -6, 0, 6, 4, -14, -56], [28, -14, 4, 2,
    ...-4, 2, 4, -14, 28], [8, 6, 4, 2, 0, -2, -4, -6, -8], [1, -1, 1, -1, 1, -1, 1, -1, 1]]
Fierz identity for commuting spinors: sum_{K in r} (a Theta_K b)(c Theta^K d) = sum_s F_rs
    ...sum_{J in s} (a Theta_J d)(c Theta^J b) : True
Hartree-Fock energy E_H + E_x = sum_s (G_s/2) D_s with G = (1 - F^T) g (random density matrix,
    ...random couplings) : True
uniform matter R = (rho_S + rho Gamma17[0])/16: only D_0 = rho_S^2, D_1 = rho^2; with g_0 = gS,
    ...g_1 = gV, g_7 = -gA, g_8 = -gP the exchange energy is that of stage 2 (section 8.7) :
    ...True
uniform density matrix: D_0 = rho_S^2, D_1 = rho^2, D_2..D_8 = 0 : True
stage-2 couplings gS = -100000, gV = 10000: G_0 = -98750, G_1 = 20000 (the HF couplings of
    ...section 8.7) : True
effective couplings G_0, G_1 in terms of all nine g_r : (15*g0/16 - g1/2 - 7*g2/4 - 7*g3/2 -
    ...35*g4/8 - 7*g5/2 - 7*g6/4 - g7/2 - g8/16, -g0/16 + 11*g1/8 - 7*g2/8 + 7*g3/8 - 7*g5/8 +
    ...7*g6/8 - 3*g7/8 + g8/16)
example g = (-100000, 10000, 3000, 0, ..., 0): G_0, G_1 : (-104000, 17375)
nonlinear field equation: dL_int/d psibar_a = -sum_r g_r sum_K (psibar Theta_K psi)(Theta^K
    ...psi)_a : True
the chirality chi17 commutes with every generator Sigma^AB, hence with every spin connection
    ...Omega_mu : True
curved chiral equations: E_L^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) +
    ...tauO.d/(a,b)) phi_L - m phi_R : True
curved chiral equations: E_R^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) -
    ...tauO.d/(a,b)) phi_R - m phi_L : True
flat limit a = b = 1, kF = kF0 (stage-2 saturation): m* = 0.3480973401..., E/A - m =
    ...-0.0283214279... : True
flat limit: isotropic pressure pa = pb = rho eps_F - eps (Gibbs-Duhem, eps_F = V0 + sqrt(kF^2 +
    ...m*^2)) : True
NUM flat: m*, rho, eps, p : (0.3480973401462447, np.float64(1.3881002132316779e-05),
    ...np.float64(1.3487872330347983e-05), np.float64(-1.588186776101813e-20))
energy-momentum conservation eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for the KS matter
    ...(arbitrary a, b; relative residual) : True
NUM conservation residual / eps' : -5.166108748755716e-12
primordial (Einstein-Kasner background, comoving kF = kF0 at x0 = 1): m*(x0) for x0 = 1e-4 ...
    ...10 : (0.003824179379430487, 0.010783989711922532, 0.030238519612042748,
    ...0.08844464116180238, 0.3480973401462447, 0.8917637493589158)
primordial: rho(x0) x0 = kF0^7/(105 pi^4) (density grows like 1/x0) : True
primordial: m* decreases toward 0 as x0 -> 0 (blue-shifted ordinary momenta, attractive scalar
    ...coupling) : True
primordial: anisotropic pressure pa/pb at x0 = 1e-3, 1e-1, 10 :
    ...(np.float64(1.0868869169150372), np.float64(1.6461737747098435),
    ...np.float64(-0.4036333858521348))
modern (static auxiliary space b = 1): m*(a) for a = 1, 2, 4, 10, 100 : (0.3480973401462447,
    ...0.853540809048541, 0.9807410717586909, 0.9987576007278518, 0.9999987563024887)
modern: eps a^3 -> constant and pa/eps -> 0 (dirac16complex behaves as pressureless matter in
```

```
    ...ordinary space) : True
modern: pb/eps at a = 100 (stress along the auxiliary space, balanced by the stabilizing
    ...stress) : 0.049773512087215956
far future (de Sitter a = e^{0.1 x0}, b = 1): m*(x0) for x0 = 0, 10, 20, 40, 80 -> m = 1 :
    ...(0.3480973401462447, 0.9395796910293632, 0.996923581335845, 0.9999923586141617,
    ...0.9999999999530483)
far future: rho = rho0 e^{-3 H x0} and 1 - m* -> 0 like rho : True
the 00 closed forms of E^(1), E^(2), E^(3) (pair-set algorithm, tools/lovelock71.py) agree with
    ...the closed forms of section 12.6 of 7+1.nb : True
Einstein-Lovelock equations are linear in (Ha', Hb') (quasi-linear second-order system) : True
Einstein + KS matter (kappa = 1000, Lambda = 0, Ha(0) = 0.05): Hb(0) from the constraint, (a,
    ...b, Ha, Hb) at x0 = 20, max relative constraint violation : (0.009143712444558836,
    ...(np.float64(1.5817386798469366), np.float64(1.1975585618423046),
    ...np.float64(0.011877999367582402), np.float64(0.007419107766685833)),
    ...np.float64(8.415965253866209e-12))
Einstein-Lovelock (alpha2 = 20, alpha3 = 500) + KS matter: Hb(0), (a, b, Ha, Hb) at x0 = 20,
    ...max relative constraint violation : (0.014980703942304356,
    ...(np.float64(1.4773514780943342), np.float64(1.276084053398673),
    ...np.float64(0.010175914492223711), np.float64(0.008488764693542736)),
    ...np.float64(4.387597609521105e-12))
the constraint Sum alpha_k E^(k)0_0 - Lambda = kappa eps is preserved by the evolution
    ...(relative violation < 1e-7) in both cases : True
boolean checks: 29   failed: 0
VERDICT: ALL CHECKS PASSED
```